

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250276854

Kind Code

A1

Publication Date

September 04, 2025

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GUARD ASSEMBLY FOR A CONVEYOR SYSTEM

Abstract

Various embodiments are directed to a guard assembly for a conveyor system. The conveyor system includes a plurality of rollers rotatably disposed between a pair of side frames of the conveyor system. The guard assembly includes a top cover and a holder such that the top cover is disposed in a gap between two adjacent rollers of the plurality of rollers of the conveyor system; the holder is detachably coupled to the top cover and is configured to secure the top cover between the two adjacent rollers of the conveyor system. The guard assembly further comprises at least one rotatable element coupled to at least one of the top cover or the holder wherein the rotatable element is configured to engage with at least one adjacent roller of the conveyor system.

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Family ID: 96880935

Appl. No.: 18/591532

Filed: February 29, 2024

Publication Classification

Int. Cl.: B65G21/20 (20060101)

U.S. Cl.:

CPC B65G21/20 (20130101);

Background/Summary

FIELD OF THE INVENTION

[0001] Various embodiments described herein relate generally to a material handling system for handling objects and, more particularly, to apparatuses for use with a conveyor system configured to facilitate transportation of objects along a conveyor surface.

BACKGROUND

[0002] Generally, in material handling environments like, but not limited to, distribution centers, warehouses, inventories, or shipping centers, a material handling system can convey, handle, sort, and organize various type of articles (for example objects, cartons, cases, containers, shipment boxes, totes, packages, and/or the like) using one or more conveyor systems. Through applied effort, ingenuity, and innovation, many of these identified problems have been solved by developing solutions that are included in embodiments of the present disclosure, many examples of which are described in detail herein.

SUMMARY

[0003] According to an embodiment of the present disclosure, a guard assembly for a conveyor system is provided. In various embodiments, the conveyor system includes a plurality of rollers rotatably disposed between a pair of side frames of the conveyor system. In various embodiments, the guard assembly includes a top cover and a holder such that the top cover is disposed in a gap between two adjacent rollers of the plurality of rollers of the conveyor system; the holder is detachably coupled to the top cover; the holder is configured to secure the top cover between the two adjacent rollers of the conveyor system. The guard assembly further includes at least one rotatable element coupled to one of the top cover or the holder wherein the rotatable element is configured to engage with at least one adjacent roller of the conveyor system.

[0004] In various embodiments, the rotatable element includes a cylinder extending along a length of the adjacent roller of the plurality of rollers. In various embodiments, a diameter of the cylinder is in a range between 30 mm to 100 mm, and a length of the cylinder is in a range between 50 mm to 3000 mm.

[0005] In various embodiments, the rotatable element is configured to rotate about an axis based on the engagement with the adjacent roller of the conveyor system.

[0006] According to another embodiment of the present disclosure, a top cover for a guard assembly for a conveyor system is provided. The top cover includes a bottom wall, first sidewall, a second sidewall and a first rotatable element coupled to one of the first sidewall or the second sidewall. The first rotatable element is configured to engage with an adjacent roller of the conveyor system. The bottom wall extends along a portion of a distance between the side frames of the conveyor system such that the bottom wall is located in a gap between adjacent rollers of the plurality of rollers. The first and the second sidewall extend from the bottom wall.

[0007] In various embodiments, the first sidewall of the top cover includes a first inclined wall extending from the bottom wall and a first flange extending substantially horizontally from the first inclined wall. The second sidewall of the top cover includes a second inclined wall extending from the bottom wall and a second flange extending substantially horizontally from the second inclined wall. The first rotatable element is coupled to one of the first flange or the second flange of the top cover.

[0008] In various embodiments, the first rotatable element includes a cylinder extending along a length of the adjacent roller of the plurality of rollers. In various embodiments, a diameter of the cylinder is in a range between 30 mm to 100 mm, and a length of the cylinder is in a range between 50 mm to 3000 mm.

[0009] In various embodiments, the first rotatable element is configured to rotate about an axis based on the engagement with the adjacent roller of the conveyor system.

[0010] In various embodiments, the top cover further includes a second rotatable element coupled to one of the first flange or the second flange of the top cover.

[0011] According to yet another embodiment of the present disclosure, a holder for a guard assembly for a conveyor system is provided. The holder includes a base wall, a first arm extending from a first side of the base wall, a second arm extending from a second side of the base wall and a third rotatable element coupled to one of the first arm or the second arm of the holder such that the third rotatable element is configured to engage with an adjacent roller of the conveyor system.

[0012] In various embodiments, the third rotatable element includes a cylinder extending along a length of the adjacent roller of the plurality of rollers. In various embodiments, a diameter of the cylinder is in a range between 30 mm to 100 mm, and a length of the cylinder is in a range between 50 mm to 3000 mm.

[0013] In various embodiments, the third rotatable element is configured to rotate about an axis based on the engagement with the adjacent roller of the conveyor system.

[0014] In various embodiments, the holder includes at least one support rib extending between the first arm and the second arm.

[0015] In various embodiments, the base wall includes a plurality of protrusions configured to prevent rotation of a fastener disposed therebetween.

[0016] In various embodiments, the holder further includes a fourth rotatable element coupled to one of the first arm or the second arm of the holder.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] The description of the illustrative embodiments can be read in conjunction with the accompanying figures. It will be appreciated that for simplicity and clarity of illustration, elements illustrated in the figures have not necessarily been drawn to scale. For example, the dimensions of some of the elements are exaggerated relative to other elements. Same numbers are used throughout the figures to reference like features and components. Embodiments incorporating teachings of the present disclosure are shown and described with respect to the figures presented herein, in which:

[0018] FIG. 1A illustrates a top view of a conveyor system, in accordance with various embodiments of the present disclosure;

[0019] FIG. 1B illustrates a bottom perspective view of the conveyor system, in accordance with various embodiments of the present disclosure;

[0020] FIG. 1C illustrates a front view of a guard assembly and two adjacent rollers of the conveyor system, in accordance with various embodiments of the present disclosure;

[0021] FIG. 2A illustrates a top perspective view of a top cover of the guard assembly, in accordance with various embodiments of the present disclosure;

[0022] FIG. 2B illustrates a bottom perspective view of the top cover of FIG. 2A, in accordance with various embodiments of the present disclosure;

[0023] FIG. 3A illustrates a top perspective view of a holder of the guard assembly, in accordance with various embodiments of the present disclosure;

[0024] FIG. 3B illustrates a bottom perspective view of the holder of FIG. 3A, in accordance with various embodiments of the present disclosure; and

[0025] FIG. 4 illustrates a method of assembling a guard assembly on a conveyor system, according to an aspect of the present disclosure.

DETAILED DESCRIPTION

[0026] Indeed, various embodiments may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Other embodiments, and modifications and improvements of the described embodiments, will occur to those skilled in the art and all such other embodiments, modifications and improvements are within the scope of the present disclosure.

Features from one embodiment or aspect may be combined with features from any other embodiment or aspect in any appropriate combination. For example, any individual or collective features of method aspects or embodiments may be applied to apparatus, product or component aspects or embodiments and vice versa. Like numbers refer to like elements throughout.

[0027] It should be understood at the outset that, although illustrative implementations of one or more aspects are illustrated below, the disclosed assemblies, systems, and methods may be implemented using any number of techniques, whether currently known or not yet in existence. The disclosure should in no way be limited to the illustrative implementations, drawings, and techniques illustrated below, but may be modified within the scope of the appended claims along with their full scope of equivalents. While values for dimensions of various elements are disclosed, the drawings may not be to scale.

[0028] The words “example,” or “exemplary,” when used herein, are intended to mean “serving as an example, instance, or illustration.” Any implementation described herein as an “example” or “exemplary embodiment” is not necessarily preferred or advantageous over other implementations. The phrases “in an embodiment,” “in some embodiments,” “according to one embodiment,” “in various embodiments” and the like generally mean that a particular feature, structure, or characteristic following the phrase may be included in at least one embodiment of the present disclosure or may be included in more than one embodiment of the present disclosure (importantly, such phrases do not necessarily refer to the same embodiment).

[0029] As used herein, directional terms used to describe a component, configuration, action, position, direction, and/or the like (for example, “rearward,” “an upward vertical direction,” “laterally outward,” “bottom,” “top,” and the like) are meant to be interpreted relative to a hypothetical conveyor system provided on an at least substantially flat horizontal surface, but the terms are not to be interpreted as requiring the conveyor system to be in this orientation at any particular time.

[0030] If the specification states a component or feature “may,” “can,” “could,” “should,” “would,” “preferably,” “possibly,” “typically,” “optionally,” “for example,” “often,” or “might” (or other such language) be included or have a characteristic, that specific component or feature is not required to be included or to have the characteristic. Such a component or feature may be optionally included in some embodiments, or it may be excluded.

[0031] Conveyor systems may be used in, for example, industrial manufacturing and packaging applications to facilitate the transportation of objects to a desired downstream location within a factory or a warehouse. For example, conveyor systems can include a plurality of cylindrical rollers disposed parallel and adjacent to each other along a length of the conveyor system. The plurality of rollers is arranged relative to one another such that the respective rolling surfaces of the rollers collectively define a conveyor path along which an object may be, at least, continuously transported towards the downstream location. Each roller can rotate about an axis thereof to help transfer objects disposed thereon to an adjacent roller and, in doing so, together, the plurality of cylindrical rollers supports in transferring the objects along a conveyor travel path towards an end of the conveyor system. Further, the conveyor systems include conveyor frame(s) having various structural components, such as, for example, but not limited to, sidewalls, panels, and/or the like, which are assembled relative to one another to define extremes capable of supporting the plurality of rollers therebetween.

[0032] In some examples, a guard apparatus may be installed within the conveyor system, by being secured relative to the side frames of the conveyor system, to prevent unauthorized and/or unintended access to an internal portion of the conveyor system, via a gap between any two adjacent rollers of the conveyor system. The guard apparatus may also prevent accumulation of debris, or small articles (such as fastener or coins) which may affect operation of the conveyor system. Further, the guard apparatus may also prevent accidental finger injury of an operator handling the objects on the conveyor system.

[0033] Additionally, the guard apparatus is subjected to repeated loading cycles, leading to localized deformations. Such deformations may cause a loosening of the guard apparatus installed within the conveyor system, which may result in undesirable movement thereof relative to the conveyor frame, undesirable noise (for example, rattling), partial inoperability of the guard apparatus, and/or unintentional disassembly of the guard apparatus relative to the conveyor system. Often, frictional force is developed between the rotating rollers and the conventional guard apparatus, resulting in reduced life and reliability thereof, undesirable loosening of the guard apparatus, and/or unintentional breakage of the guard apparatus. The conventional guard apparatus is known to include complicated profiles configured to address issues resulting from the development of the friction as mentioned herein. However, such complicated profiles of the guard apparatus increase design complexity besides an increase in cost of manufacturing.

[0034] Various embodiments of the present disclosure are directed to a guard assembly for a conveyor system. The conveyor system includes a plurality of rollers rotatably disposed between a pair of side frames of the conveyor system. In various embodiments, the guard assembly includes a top cover and a holder such that the top cover is disposed in a gap between two adjacent rollers of the plurality of rollers of the conveyor system. The holder is detachably coupled to the top cover and is configured to secure the top cover between the two adjacent rollers of the conveyor system. The guard assembly further includes at least one rotatable element coupled to one of the top cover or the holder wherein the rotatable element is configured to engage with at least one adjacent roller of the conveyor system.

[0035] FIG. 1A illustrates a top view of a conveyor system **100**, in accordance with various embodiments of the present disclosure. The conveyor system **100** includes a plurality of rollers **102** (such as a first roller **102-1**, a second roller **102-2**, a third roller **102-3**, and a fourth roller **102-4**), hereinafter alternatively, individually, and collectively referred to as “the roller(s) **102**” or “adjacent rollers **102-1**, **102-2**”, disposed between a pair of side frames **104-1**, **104-2** (hereinafter alternatively and collectively referred to as “the side frames **104**”), such that a gap **106** exists between any two adjacent rollers **102-1**, **102-2**. For example, the gap **106** exists between two adjacent rollers **102-1**, **102-2** or **102-3**, **102-4**. In various embodiments, the adjacent rollers **102-1**, **102-2** are joined by one or more driving members **103** such as a belt or an O-ring, at one end, where the driving member **103** is configured to transmit a rotary motion from one roller to the other, without slippage.

[0036] The conveyor system **100** may include multiple such driving members, such as **103-1**, **103-2** and so on, configured to connect each pair of adjacent rollers to transmit the motion throughout a length of the conveyor system **100** in order to facilitate transfer of objects. As seen in FIG. 1, a guard assembly **108** is positioned within the conveyor system **100** to provide coverage over at least a portion of the gap **106** between the two adjacent rollers **102-1**, **102-2**. In some embodiments, the guard assembly **108** provides coverage over an entire portion of the gap **106** between the side frames **104-1** and **104-2**.

[0037] Referring to FIG. 1B, a bottom perspective view of the conveyor system **100** is illustrated. FIG. 1B is described in conjunction with FIG. 1A. The guard assembly **108** includes a top cover **110** and a holder **112**. The top cover **110** is disposed in the gap **106** between two adjacent rollers **102-1**, **102-2** of the plurality of rollers **102**. The holder **112** is detachably coupled to the top cover **110** and configured to secure the top cover **110** between the two adjacent rollers **102-1**, **102-2**. In various embodiments, the top cover **110** extends between the side frame **104-1**, **104-2** and is configured to provide coverage over the entire portion of gap **106** between the side frames **104-1**, **104-2**.

[0038] Referring to FIG. 1C, a front view of the guard assembly **108** and the two adjacent rollers **102-1**, **102-2** of the conveyor system **100** is illustrated. FIG. 1C is described in conjunction with FIG. 1A and FIG. 1B. As stated above, the holder **112** is detachably coupled to the top cover **110** and configured to secure the top cover **110** between the two adjacent rollers **102-1**, **102-2**. In some

embodiments, a first fastener **105**, such as a screw or a bolt, defining a head **105-1** and a shank **105-2** is used in combination with a second fastener **107**, such as a nut, to secure the top cover **110** and the holder **112**. In some embodiments, the top cover **110** is disposed below the driving member **103-1**. As such, a top surface of the top cover **110** is located at a distance from the driving member **103-1**, thereby preventing potential friction development between the top cover **110** and the driving member **103-1**. In some embodiments, the guard assembly **108** includes at least one rotatable element **114** coupled to at least one of the top cover **110** or the holder **112**. For example, the guard assembly **108** includes a plurality of rotatable elements **114** (such as a first rotatable element **114-1**, a second rotatable element **114-2**, a third rotatable element **114-3** and a fourth rotatable element **114-4**), hereinafter alternatively, individually, and collectively referred to as “the rotatable element(s) **114**”.

[0039] In an installed condition of the guard assembly **108**, the rotatable element **114** is configured to engage with at least one adjacent roller **102-1**, **102-2** of the conveyor system **100**. As used hereinafter, the term “installed condition” refers to the conveyor system **100** including the guard assembly **108** disposed in the gap **106** between the adjacent rollers **102-1** and **102-2**, as illustrated in FIG. 1C. In various embodiments, the rotatable element **114** defines a diameter “D” and a length “L”. In the illustrated embodiments, the rotatable element(s) **114** is implemented as a cylinder extending along a length of the adjacent roller **102-1**, **102-2** of the plurality of rollers **102**.

[0040] In an operating condition of the conveyor system **100**, to transfer one or more object along a length direction **122** of the conveyor system **100**, the adjacent rollers **102-1** and **102-2** rotates in a first direction **124**, as shown in FIG. 1C. As used hereinafter, the term “operating condition” refers to the conveyor system **100** with the plurality of rollers **102** rotating about each of their axis in order to transfer the one or more objects.

[0041] As shown in FIG. 1C, the top cover **110** includes the rotatable elements **114-1** and **114-2** coupled to either side thereof and configured to engage with the adjacent rollers **102-1** and **102-2**, respectively, in the installed condition. Similarly, the holder **112** includes the rotatable elements **114-3** and **114-4** coupled to either side thereof and configured to engage with the adjacent rollers **102-1** and **102-2**, respectively, in the installed condition. As shown in FIG. 1C, in the installed condition, the diameter of the rotatable elements **114-1**, **114-2**, **114-3**, **114-4** determines the position of the top cover **110** with respect to the adjacent rollers **102-1**, **102-2**. For example, as may be observed in FIG. 1C, with an increase in diameter of the rotatable elements **114-1** and **114-2**, the top cover **110** may be displaced in a vertical direction with respect to the adjacent rollers **102-1** and **102-2**. However, the diameter of the rotatable elements **114-1**, **114-2**, **114-3**, **114-4** may be pre-determined to prevent contact between the top cover **110** and the adjacent rollers **102-1**, **102-2**, and between the top cover **110** and the driving member **103-1**. This way, during the operating condition, the frictional forces generated between the top cover **110** and the adjacent rollers **102-1**, **102-2** may be eliminated. In various embodiments, the diameter “D” of the cylinder is in a range of about 30 mm to 100 mm, and the length “L” of the cylinder is in a range of about 50 mm to 3000 mm.

[0042] In various embodiments, in the operating condition, the rotatable elements **114-1** and **114-3** engage with the adjacent roller **102-1** and rotate about their respective axis in a second direction **126** opposite to the first direction **124**, the rotatable elements **114-2** and **114-4** engage with the adjacent roller **102-2** and rotate about their respective axis in the second direction **126**. As the rotatable elements **114-1**, **114-2**, **114-3** and **114-4** rotate along with the adjacent rollers **102-1**, **102-2** in said opposing directions, a relative motion between the rotatable elements **114-1**, **114-2**, **114-3**, **114-4** and the plurality of rollers **102** is averted, thereby minimizing the friction generated due to the contact therebetween.

[0043] Shape of the rotatable element **114** should not be construed as limited. Instead, other shapes and features will be apparent to a person skilled in the art, albeit with few changes to the features described and illustrated herein. For example, in some embodiments, the rotatable element **114** may be implemented as one of a sphere or a cone. Further, the rotatable element **114** may include

multiple fillets, chamfer, or edge radii. In various embodiments, the rotatable element **114** may include plurality of hemi-spherical protrusions configured to engage with the adjacent rollers **102-1**, **102-2**. The hemi-spherical protrusions may be laterally spaced apart from each other and further reduce the contact area of the rotatable element **114** with the adjacent rollers **102-2**, **102-2**, thereby further reducing the friction generated therebetween. In various embodiments, the length “L” of the rotatable element **114** may be varied to minimize the contact area between the rotatable element(s) **114** and the plurality of rollers **102**.

[0044] FIG. 2A illustrates a top perspective view of the top cover **110** and FIG. 2B illustrates a bottom perspective view of the top cover **110**. The top cover **110** includes a bottom wall **202**, a first sidewall **204** located on one side of the bottom wall **202**, a second sidewall **206** located on an opposite side of the bottom wall **202**, the first rotatable element **114-1** coupled to the first sidewall **204** of the top cover **110**, and the second rotatable element **114-2** coupled to the second sidewall **206** of the top cover **110**. The bottom wall **202** extends along a portion of a distance between the side frames **104-1**, **104-2** of the conveyor system **100**. The bottom wall **202** is configured to be located in the gap **106** between the adjacent rollers **102-1**, **102-2**. In various embodiments, the top cover **110** may be made of light-weight polymer, such as plastic, and may be manufactured through, for example, injection molding. To this end, the bottom wall **202**, the first sidewall **204** and the second sidewall **206** are integrally molded to form the top cover **110**. The bottom wall **202** may also define a receptacle (not shown) configured to receive the head **105-1** of the first fastener **105** therein. The bottom wall **202** may define a first hole **203** (see FIG. 2B) configured to receive the shank **105-2** of the first fastener **105** therein.

[0045] In various embodiments, the first sidewall **204** of the top cover **110** includes a first inclined wall **214** located on one side of the bottom wall **202** and a first flange **216** extending substantially horizontally from the first inclined wall **214**. The second sidewall **206** of the top cover **110** includes a second inclined wall **218** located on an opposite side of the bottom wall **202** and a second flange **220** extending substantially horizontally from the second inclined wall **218**. The first flange and **216** and the second flange **220** extend in opposite directions from the first inclined wall **214** and the second inclined wall **218**, respectively. In some embodiments, the first rotatable element **114-1** is coupled to the first flange **216** of the top cover **110**. In various embodiments, the first rotatable element **114-1** is configured to rotate about a first axis based on the engagement with the adjacent roller **102-1**, **102-2** during operating condition of the conveyor system **100**. In various embodiments, the top cover **110** further includes the second rotatable element **114-2** coupled to the second flange **220** of the top cover **110**.

[0046] In various embodiments, the rotatable elements **114** may be disposed on either side of the top cover **110** and may be laterally spaced apart from each other. In various embodiments, a length of the top cover **110** may determine the number of rotatable elements **114** coupled thereto. To this end, the rotatable elements **114** may be uniformly disposed at pre-determined distance from each other along a length of the top cover **110**. In various embodiments, the length “L” of each rotatable element **114** may be varied to vary the contact area between the rotatable element(s) **114** and the corresponding roller **102**. As will be understood from the FIG. 2A read in light of FIG. 1C, increase in length of the rotatable element **114** causes an increase in area of contact with the corresponding roller **102** and, accordingly, a decrease in length of the rotatable element **114** causes a decrease in area of contact with the corresponding roller **102**. As such, the number of rotatable element(s) **114** to be coupled to the top cover **110** may be determined based on the length of the top cover **110**. In some embodiments, a length of one rotatable element **114** may be greater than other rotatable elements **114** coupled to the top cover **110**. For example, the top cover **110** may include three rotatable elements **114** and a middle rotatable element **114** may have a length greater than lengths of the other two rotatable elements **114**. In this embodiment, distances between the three rotatable elements **114** may be non-equal. In cases where the rollers **102** are tapered, such non-equal distances between the rotatable elements **114** along the top cover **110** may be implemented.

Embodiments disclosed herein with respect to configuration of the rotatable elements **114** and their placement on the top cover **110** should not be construed as limited. Multiple such variations will be apparent to a person skilled in the art, based on the description herein, albeit with few variations. [0047] Referring to FIG. 2B, in various embodiments, the top cover **110** may include a first projection **228** and a second projection **230** extending from the first sidewall **204** thereof, where the first projection **228** and the second projection **230** are parallel to each other. With aid of a first supporting pin (not shown) that extends between the first projection **228** and the second projections **230**, the rotatable element **114-2** may be rotatably supported therebetween.

[0048] Shape of the illustrated top cover **110** should not be construed as limited. Instead, other shapes and features will be apparent to the person skilled in the art, albeit with few changes to the features described and illustrated herein. For example, the top cover **110** may include a U-shaped elongated wall with extending side flanges. Further, the top cover **110** may include multiple fillets, chamfer, or edge radii.

[0049] Referring to FIG. 3A, a top perspective view of the holder **112** is illustrated. The holder **112** includes a base wall **302**, a first arm **304** extending from a first side **306** of the base wall **302**, a second arm **308** extending from a second side **310** of the base wall **302**, and the third rotatable element **114-3** (See FIG. 1C) coupled to the first arm **304** of the holder **112**, such that the third rotatable element **114-3** is configured to engage with an adjacent roller **102-1** of the conveyor system **100**.

[0050] Referring to FIG. 3B, a bottom perspective view of the holder **112** is illustrated. FIG. 3B is described in conjunction with FIG. 1A through FIG. 3A. In various embodiments, the third rotatable element **114-3** is configured to rotate about a third axis **328** based on the engagement with the adjacent roller **102-1** of the conveyor system **100**. In various embodiments, the holder **112** includes at least one support rib **314** extending between the first arm **304** and the second arm **308**. In various embodiments, the base wall **302** includes a plurality of protrusions **316** configured to prevent rotation of the second fastener **107** disposed therebetween. For example, the plurality of protrusions **316** engages, through an interference fit, with side surfaces (not shown) of the second fastener **107**, such that the rotational motion of second fastener **107** is restricted. Further, the base wall **302** defines a second hole **303** configured to receive the shank **105-2** of the first fastener **105** therethrough. In various embodiments, the holder **112** further includes a fourth rotatable element **114-4** coupled to the second arm **308** thereof. In various embodiments, with the aid of a second supporting pin, the third rotatable element **114-3** is rotatably supported on the first arm **304**. In some embodiments, each of the first arm **304** and the second arm **308** may include two or more rotatable elements in lieu of single rotatable element as illustrated in FIG. 3A and FIG. 3B.

[0051] In some embodiments, the first arm **304** of the holder **112** may include a first inclined portion **305** extending from the first side **306** of the base wall **302** and a second inclined portion **307** extending further divergently from the first inclined portion **305**. In the illustrated embodiment, the first inclined portion **305** and the second inclined portion **307** together constitutes the first arm **304**. Similarly, the second arm **308** of the holder **112** may include a third inclined portion **309** extending from the second side **310** of the base wall **302** and a fourth inclined portion **311**, extending further divergently from the third inclined portion **309**. In the illustrated embodiment, the third inclined portion **309** and the fourth inclined portion **311** together constitutes the second arm **308**. In some embodiments, each of the first arm **304** and the second arm **308** may include multiple inclines portions. In some embodiments, the each of the first arm **304** and the second arm **308** may be implemented as arcuate portions extending respectively from the first side **306** and the second side **310** of the base wall **302**. In some embodiments, the holder **112** may have a bell-shape. In various embodiments, the third rotatable element **114-3** is coupled to the second inclined portion **307** of the holder **112** and the fourth rotatable element **114-4** is coupled to the fourth inclined portion **311** of the holder **112**.

[0052] In various embodiments, the holder **112** may be made of light-weight polymer, such as

plastic, and may be manufactured through, for example, injection molding. To this end, the support rib **314** and the plurality protrusions **316** may be integrally formed together with the base wall **302**, the first arm **304**, and the second arm **308**. In various embodiments, the holder **112** defines a hollow space “H” surrounded by the first arm **304**, the second arm **308**, and one or more support ribs **314** joining the first arm **304** and the second arm **308**. Shape of the illustrated holder **112** should not be construed as limited. Instead, other shapes and features will be apparent to a person skilled in the art, albeit with few changes to the features described and illustrated herein. For example, in some embodiments, the holder **112** may include multiple fillets, chamfer, or edge radii.

[0053] The present disclosure provides an improved guard assembly for the conveyor system **100**. As described above, the guard assembly **108** reduces the frictional forces between the stationary top cover **110** and the rotating rollers, owing to the rotatable elements **114** disposed thereon.

Further, since the top cover **110** provides coverage over the entire portion of the gap **106** between the side frames **104-1**, **104-2**, the guard assembly **108** prevents unauthorized and/or unintended access to internal portion(s) of conveyor system **100**. This is achieved by, for example, The guard assembly **108** also prevents accumulation of debris, or small article, such as fasteners or coins, which may lead to jamming of the conveyor system **100**. Further, the guard assembly **108** prevents finger injury by effectively, in a way, blocking the gap **106** between adjacent rollers **102-1**, **102-2**.

[0054] There are further advantages of the present disclosure. For example, in case of damage or deformation of any one of the rotatable elements occurring during usable life of the guard assembly **108**, the need to replace the entire guard assembly **108** may be eliminated. For example, if a single rotatable element is damaged, only the rotatable element could be replaced instead of the entire top cover **110** or the holder **112**, thereby minimizing the replacement costs.

[0055] Advantageously, in an event where a pull force is applied to remove or dismantle the guard assembly **108** from the conveyor system **100** in a substantially vertical direction with respect to the rollers **102**, the guard assembly **108** provides an effective resistance against the pull force through vertical securing thereof. For example, the rotatable elements **114** may be configured to couple with the first arm **304** and the second arm **308** of the holder **112** at their respective distal edges, such that the rotatable elements **114** engage with the adjacent rollers **102-1**, **102-2** of the conveyor system **100**, thereby vertically securing the guard assembly **108** to preventing the guard assembly **108** from coming off due to an applied pull force.

[0056] Further, the plurality of protrusions **316** on the base wall **302** of the holder **112** prevent rotation of the second fastener **107** disposed therein, thereby eliminating possibility of loosening out of the top cover **110** due to repeated loading cycles.

[0057] There are further advantages of the present disclosure. For example, the rotatable elements **114** may be configured to couple with the first flange **216** of the first sidewall **204** of the top cover **110** and the second flange **220** of the second sidewall **206** of the top cover **110** at their respective distal ends, such that the rotatable elements **114** engage with the adjacent rollers **102-1**, **102-2** of the conveyor system **100**. This engagement provides a vertical securing of the top cover **110** as they prevent the top cover **110** from falling into the gap **106** between the adjacent rollers **102-1**, **102-2**.

[0058] Advantageously, the support rib **314** adds to the strength and the stiffness of the holder **112**, thereby reducing localized deformations of the holder **112**.

[0059] There are further advantages provided by the present disclosure. It may be desirable to manufacture the top cover **110** and the holder **112** using simpler methods, such as injection molding. The present disclosure provides a simplified structure for each of the top cover **110** and the holder **112**, thereby reducing the manufacturing costs, etc. For example, the first inclined wall **214** of the first sidewall **204** and the second inclined wall **218** of the second sidewall **206** extend divergently from the bottom wall **202**. Such an embodiment ensures that there is, for example, sufficient ejection or draft angle clearance in the die used for molding the component through injection molding. Moreover, the die can have a simpler profile which reduces complex man hour effort to build the die for injection molding. Thus, it is easier to manufacture the top cover **110**

through an injection molding process as an integrated component.

[0060] Various aspects of the present disclosure provide a method of assembling the guard assembly **108** on the conveyor system **100**. FIG. **4** illustrates a method **400** of assembling the guard assembly **108** on the conveyor system **100**. FIG. **4** is described in conjunction with FIG. **1A** through FIG. **3B**. At step **402**, the method **400** includes placing the top cover **110**, from above the conveyor system **100**, such that the bottom wall **202** of the top cover **110** is disposed in the gap **106** between two adjacent rollers **102-1**, **102-2** of the plurality of rollers **102** of the conveyor system **100**. At step **404**, the method **400** includes locating the holder **112** below the top cover **110**, from beneath the conveyor system **100**, such that the first hole **203** on the bottom wall **202** of the top cover **110** aligns with the second hole **303** on the base wall **302** of the holder **112**. At step **406**, the method **400** includes inserting the first fastener **105**, such as a screw or a bolt, through the first hole **203** of the top cover **110** and the second hole **303** of the holder **112**. At step **408**, the method **400** includes tightening the first fastener **105** against the second fastener **107**, such as a nut, so that the top cover **110** and the holder **112** are securely held together between the pair of side frames **104-1**, **104-2**.

[0061] The foregoing method descriptions are provided merely as illustrative examples and are not intended to require or imply that the steps of the various embodiments must be performed in the order presented. As will be appreciated by one of skilled in the art, the order of steps in the foregoing embodiments may be performed in any order. Any reference to claim elements in the singular, for example, using the articles “a,” “an” or “the” is not to be construed as limiting the element to the singular.

[0062] When an element or layer is referred to as being “on,” “engaged to,” “connected to,” or “coupled to” another element or layer, it may be directly on, engaged, connected, or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly engaged to,” “directly connected to,” or “directly coupled to” another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (for example, “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.). As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0063] Spatially relative terms, such as “inner,” “outer,” “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. Spatially relative terms may be intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures.

[0064] Many modifications and other embodiments of the present disclosure will be apparent to one skilled in the art to which the present disclosure pertain having the benefit of teachings presented in the foregoing descriptions and the associated drawings. Although the figures only show certain components of the system described herein, it is understood that various other components may be present. Therefore, it is to be understood that the present disclosure should not be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, the steps in the method described above may not necessarily occur in the order depicted in the accompanying diagrams, and in some cases one or more of the steps depicted may occur substantially simultaneously, or additional steps may be involved. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

- 1.** A guard assembly for a conveyor system comprising a plurality of rollers rotatably disposed between a pair of side frames thereof, the guard assembly comprising: a top cover disposed in a gap between two adjacent rollers of the plurality of rollers; and a holder detachably coupled to the top cover, wherein the holder is configured to secure the top cover between the two adjacent rollers of the conveyor system; and at least one rotatable element coupled to at least one of the top cover or the holder, wherein the at least one rotatable element is configured to engage with at least one adjacent roller of the conveyor system.
- 2.** The guard assembly of claim 1, wherein the least one rotatable element comprises a cylinder extending along a length of the adjacent roller of the plurality of rollers.
- 3.** The guard assembly of claim 2, wherein a diameter of the cylinder is in a range between 30 mm to 100 mm, and a length of the cylinder is in a range between 50 mm to 3000 mm.
- 4.** The guard assembly of claim 1, wherein the least one rotatable element is configured to rotate about an axis based on the engagement with the adjacent roller of the conveyor system.
- 5.** A top cover for a guard assembly for a conveyor system comprising a plurality of rollers rotatably disposed between a pair of side frames thereof, the top cover comprising: a bottom wall extending along a portion of a distance between the side frames of the conveyor system, wherein the bottom wall is located in a gap between adjacent rollers of the plurality of rollers; a first sidewall extending from the bottom wall; a second sidewall extending from the bottom wall; and a first rotatable element coupled to one of the first sidewall or the second sidewall of the top cover, wherein the first rotatable element is configured to engage with an adjacent roller of the conveyor system.
- 6.** The top cover of claim 5, wherein: the first sidewall comprises: a first inclined wall extending from the bottom wall; and a first flange extending substantially horizontally from the first inclined wall; and the second sidewall comprises: a second inclined wall extending from the bottom wall; and a second flange extending substantially horizontally from the second inclined wall, wherein the first rotatable element is coupled to one of the first flange or the second flange of the top cover.
- 7.** The top cover of claim 5, wherein the first rotatable element comprises a cylinder extending along a length of the adjacent roller of the plurality of rollers.
- 8.** The top cover of claim 7, wherein a diameter of the cylinder is in a range between 30 mm to 100 mm, and a length of the cylinder is in a range between 50 mm to 3000 mm.
- 9.** The top cover of claim 5, wherein the first rotatable element is configured to rotate about a first axis based on the engagement with the adjacent roller of the conveyor system.
- 10.** The top cover of claim 5, further comprising a second rotatable element coupled to one of the first flange or the second flange of the top cover.
- 11.** A holder for a guard assembly for a conveyor system comprising a plurality of rollers rotatably disposed between a pair of side frames thereof, the holder comprising: a base wall; a first arm extending from a first side of the base wall; a second arm extending from a second side of the base wall; and a third rotatable element coupled to one of the first arm or the second arm of the holder, wherein the third rotatable element is configured to engage with an adjacent roller of the conveyor system.
- 12.** The holder of claim 11, wherein the third rotatable element comprises a cylinder extending along a length of the adjacent roller of the plurality of rollers.
- 13.** The holder of claim 12, wherein a diameter of the cylinder is in a range between 30 mm to 100 mm, and a length of the cylinder is in a range between 50 mm to 3000 mm.
- 14.** The holder of claim 11, wherein the third rotatable element rotates about a second axis upon engagement with the adjacent roller of the conveyor system.
- 15.** The holder of claim 11, further comprising at least one support rib extending between the first arm and the second arm.
- 16.** The holder of claim 11, wherein the base wall comprises a plurality of protrusions configured to

prevent rotation of a fastener disposed therebetween.

17. The holder of claim 11 further comprising a fourth rotatable element coupled to one of the first arm or the second arm of the holder.
